CS240 Project Proposal

Virtual Camera Sampling for Novel-View Synthesis
via Greedy Submodular Maximization

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1 Topic Selection

Self-proposed topic (related to Reference Topic #11 "Set Cover" and #2 "Graph Partitioning"). We study **virtual-camera sampling path planning** – the problem of automatically choosing where to place additional "virtual" cameras around a 3D scene reconstructed from sparse real photos, in order to most effectively improve a downstream 3D Gaussian Splatting (3DGS) renderer. The application background is **novel-view synthesis (NVS)** for AR/VR, robotics and aerial mapping: real captures are expensive and uneven, so practitioners synthesize extra training views to fill in the under-observed regions. The interesting algorithmic question is *which* virtual views to render under a tight budget – a problem that, despite its modern ML setting, reduces almost exactly to the classical **Maximum Coverage / Submodular Maximization** problem solvable by the greedy approximation algorithm with the celebrated $(1 - 1/e)$ guarantee (Nemhauser–Wolsey–Fisher, 1978).

2 Problem Formulation

Let $\mathcal{G} = \{g_1, \ldots, g_M\}$ be a trained 3DGS scene of M Gaussian primitives, $\mathcal{C}_{\text{train}}$ the set of existing real cameras, and $\mathcal{P} = \{p_1, \ldots, p_K\}$ a candidate-pose pool generated around an anchor set (translations, rotations, orbits). For each candidate camera p_i we define a coverage set

$$S_i \;=\; \big\{(g_j, d_q) \,:\, g_j \text{ visible from } p_i \text{ along discretized view direction } d_q\big\} \;\subseteq\; \mathcal{U},$$

where $\mathcal{U}$ is the universe of all (Gaussian, view-direction) pairs.

Input: $(\mathcal{G}, \mathcal{C}_{\text{train}}, \mathcal{P}, B)$ with budget $B \le K$. **Output:** a subset $\mathcal{V} \subseteq \mathcal{P}$, $|\mathcal{V}| = B$, maximising

$$f(\mathcal{V}) \;=\; \Big|\bigcup_{p_i \in \mathcal{V}} S_i \;\setminus\; \mathrm{Cov}(\mathcal{C}_{\text{train}})\Big|.$$

f is non-negative, monotone and **submodular** (coverage function), so the problem is NP-hard but admits a $(1 - 1/e)$-approximation by greedy. An alternative scoring reformulates f as the **Fisher information gain** $f_{\text{Fisher}}(\mathcal{V}) = \sum_{p_i \in \mathcal{V}} \mathrm{tr}\big(H_{p_i}\,(H_{\text{train}} + \lambda I)^{-1}\big)$, related to D-optimal experimental design.

3 Related Work

- **Selected paper for reproduction – ExploreGS** (Zu et al., arXiv:2508.06014, 2025). Proposes a virtual-camera sampling stage that grows trajectories via DFS-style greedy expansion under three policies (Fisher / Coverage / Coverage-Refine). Code: `scene/space_search.py` and `train_stage_vcam.py`.
- **FisherRF** (Jiang et al., arXiv:2311.17874, NeurIPS 2024). The canonical "active view selection via Fisher information" baseline; greedy adds one view at a time maximising $\mathrm{tr}(H_{\text{cand}} \cdot I_{\text{train}})$. Code: `active/H_reg.py`.
- **Foundational theory.** Nemhauser, Wolsey & Fisher (1978) – greedy on monotone submodular functions achieves $(1 - 1/e)$. Minoux (1978) – accelerated lazy-greedy. Mirzasoleiman et al. (2015) – stochastic greedy in linear time.

4 Algorithm and Technical Plan

Core algorithm – greedy submodular maximisation for view selection:

Algorithm 1 Greedy Virtual-Camera Selection

Require: Scene $\mathcal{G}$, candidate pool $\mathcal{P}$ ($|\mathcal{P}| = K$), budget B, scoring function f

$\mathcal{V} \leftarrow \emptyset$; $\quad \textsc{covered} \leftarrow \mathrm{Cov}(\mathcal{C}_{\text{train}})$

for $t = 1, \ldots, B$ **do**

$\quad p^* \leftarrow \arg\max_{p \in \mathcal{P} \setminus \mathcal{V}} \ \Delta f(p \,|\, \mathcal{V}) \ = \ |S_p \setminus \textsc{covered}|$ $\quad \triangleright$ marginal gain

$\quad \mathcal{V} \leftarrow \mathcal{V} \cup \{p^*\}$; $\quad \textsc{covered} \leftarrow \textsc{covered} \cup S_{p^*}$

end for

return $\mathcal{V}$

Complexity (to be analysed rigorously in the report). Naive cost is $\mathcal{O}(B \cdot K \cdot M \cdot Q)$ where M is the number of Gaussians and Q the number of view-direction bins; with **lazy evaluation** (Minoux 1978) the per-round work drops sharply in practice. The Fisher variant costs $\mathcal{O}(B \cdot K \cdot D)$ with D the number of 3DGS parameters; we will analyse the trade-off between coverage (cheap, geometric) and Fisher (expensive, learning-aware).

Implementation plan.

1. Reuse ExploreGS's voxel/visibility infrastructure (`occupancy_grid`, `grid_view_dirs`) and FisherRF's Hessian estimator.
2. Extract the standalone greedy loop from ExploreGS's DFS-trajectory wrapper, isolating the pure submodular setting for analysis.
3. Implement **lazy-greedy** as an optimisation comparison.
4. **Datasets:** Mip-NeRF 360 (*bicycle*, *garden*) for object-centric scenes and Tanks & Temples (*truck*, *train*) for scene-level evaluation – both already supported in the ExploreGS pipeline.
5. **Evaluation:** for a fixed real-camera split, run each policy to select $B \in \{5, 10, 20, 40\}$ virtual cameras, retrain a 3DGS for 30k iters, and report PSNR / SSIM / LPIPS on a held-out test set.

5 Project Scope and Expected Goals

Reproduction (Stages 1–3). Reproduce the Coverage-Greedy policy from ExploreGS Table 2 on Mip-NeRF 360, and the Fisher policy from FisherRF Table 1 on the same scenes.

Experimental Evaluation (Stage 4).

- *Runtime / scalability*: measure per-round greedy cost as K grows from 50 to 5000; demonstrate the $\mathcal{O}(BK)$ scaling and lazy-greedy speedup.
- *Quality vs. budget curve*: PSNR vs. B for {Coverage-Greedy, Fisher-Greedy, Random, Farthest-Point}.
- *Approximation quality*: for small instances, compare $f(\mathcal{V}_{\text{greedy}})$ against an ILP-computed optimum to empirically verify that the $(1 - 1/e)$ bound is loose in practice.

Optional Extension (Stage 5). Implement **lazy-greedy** (Minoux) and **stochastic-greedy** (Mirzasoleiman et al. 2015), comparing wall-clock against plain greedy; evaluate on one extra dataset (a UAV scene) to test out-of-distribution scalability.

6 Team Information

Member	Student ID	Planned Contribution
Jiyuan Cao (team lead)	2025133812	Greedy core + lazy-greedy implementation, complexity analysis, experiments on Mip-NeRF 360
Zhongyao Tuo	2025232116	Coverage scoring (visibility / voxel grid), dataset preparation, evaluation pipeline (PSNR/SSIM/LPIPS)
Beiyi Zhu	2025233244	Fisher scoring port from FisherRF, Hessian profiling, runtime/scalability benchmarks, write-up and figures